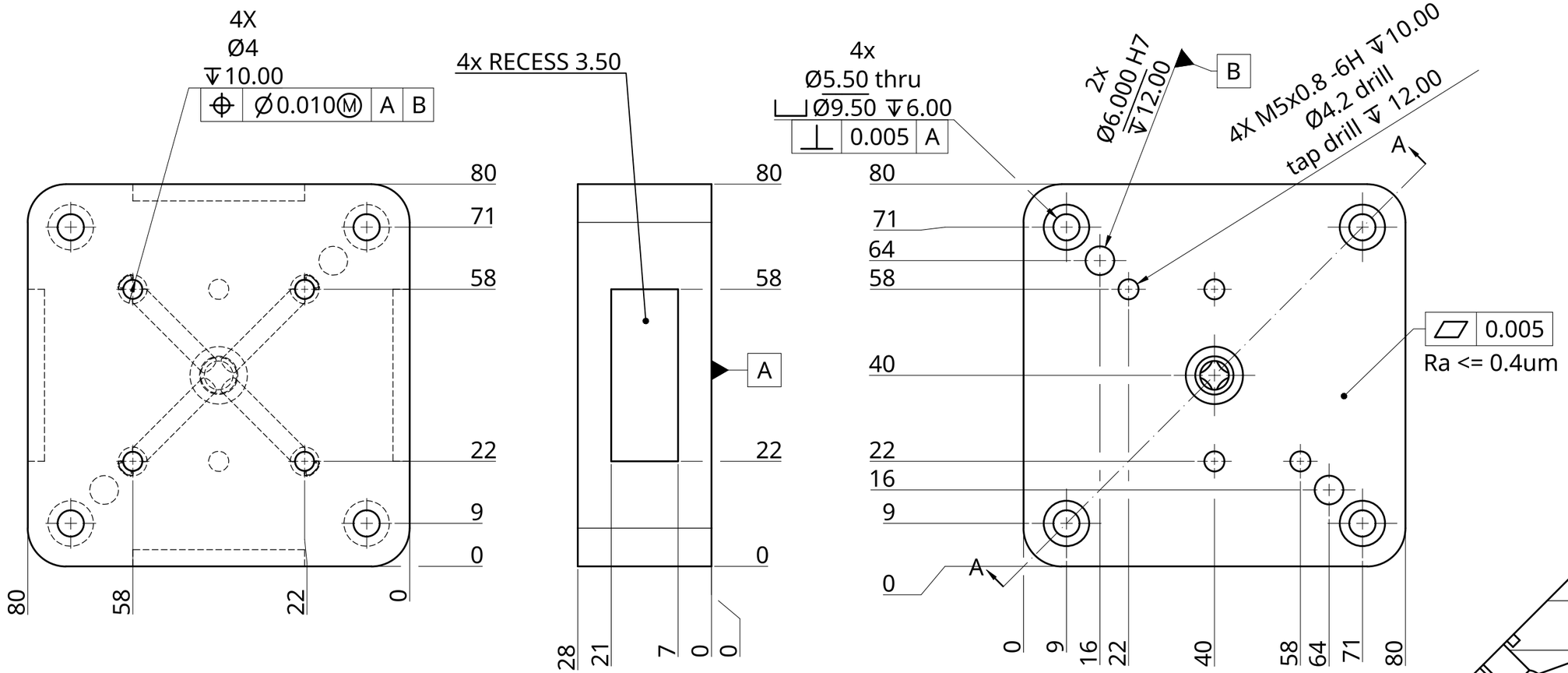


NOTES: UNLESS OTHERWISE SPECIFIED

1. GENERAL & STANDARDS:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (WUL_MANIFOLD_RevA.STEP) GOVERNS NOMINAL GEOMETRY.
   1.3 GENERAL TOLERANCES: LINEAR +/- 0.1 mm,
ANGULAR +/- 0.5 DEG (PER ISO 2768-m).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
2. MATERIAL & THERMAL POST-PROCESSING:
   2.1 PRIMARY MATERIAL: ADDITIVE GRADE AlSi10Mg PER ASTM F3318 (PBF-LB / SLM).
   2.2 VACUUM STRESS RELIEF: 320°C +/- 5°C FOR 120 MINS IN INERT VACUUM FURNACE.
   2.3 CALCULATED NET MASS: 342.8 g (VOLUME: 128,872 mm^3).
3. SURFACE TREATMENT & PASSIVATION:
   3.1 HARD ANODIZING: HIGH-PURITY SEMICONDUCTOR HARD ANODIZATION PER MIL-A-8625 TYPE III.
   3.2 THICKNESS: 35 +/- 5 MICRONS. SEALING: HIGH-PURITY DEIONIZED WATER SEAL.
   3.3 SEALING FACE ROUGHNESS: Ra <= 0.4 MICRONS; NON-SEALING FACES: Ra <= 1.6 MICRONS.
4. CLEANLINESS & VACUUM CRITERIA:
   4.1 NO OIL, GREASE, RESIDUAL CUTTING FLUIDS, OR AIRBORNE PARTICLES PERMITTED.
   4.2 COMPONENT MUST UNDERGO PRECISION ULTRASONIC CLEANING USING REAGENT-GRADE IPA FOLLOWED BY HIGH-PURITY DEIONIZED (DI) WATER RINSE (>18 MEGOHM-CM).
   4.3 VACUUM LEAK RATE: GLOBAL HELIUM MASS SPECTROMETER LEAK RATE MUST BE LESS THAN 1.0E-10 mbar*L/s PER ASTM E498.
5. CLEANROOM HANDLING & PACKAGING:
   5.1 FINAL DRYING, INSPECTION, AND PACKAGING SHALL BE PERFORMED IN AN ISO CLASS 5 (FED-STD-209E CLASS 100) CLEANROOM ENVIRONMENT.
   5.2 DOUBLE-BAG SEALED: INNER AND OUTER BAGS SHALL BE CLEANROOM-CERTIFIED ANTISTATIC POLYETHYLENE (PE), HEAT-SEALED UNDER HIGH-PURITY NITROGEN (N2) PURGE.
   5.3 EACH PACKAGE MUST BE CLEARLY LABELED WITH DWG NO, LOT NO, AND CLEANROOM DATE.

		NAME	SIGNATURE	DATE
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN MILLIMETERS	DRAWN			2026-10-11
ANGULAR = ± °	CHECKED			
SURFACE FINISH	APPROVED			
DO NOT SCALE DRAWING				
BREAK ALL SHARP EDGES AND REMOVE BURRS				
THIRD ANGLE PROJECTION	MATERIAL: AlSi10Mg per ASTM F3318		FINISH	

TITLE: SEMICONDUCTOR ETCH CHAMBER PROCESS GAS DISTRIBUTION MANIFOLD

SIZE	DWG NO.	REV.
A3	WUL_MANIFOLD_REVA	A

SCALE	WEIGHT	SHEET
NTS	0.343 kg	1 of 1